Problems with Iterated Elimination of dominated strategies

- Common knowledge that players are rational at each step. All of them know that all players are rational. All players know that all the players know that all the players know that all the players are rational and so on.
- Imprecise solution

Nash equilibrium:

Definition: In the n-player normal form game $G = \{S_1, S_2 ... S_n; u_1, u_2 ... u_n\}$, the strategies $(s_1^*, s_2^*, s_3^* ...., s_n^*)$ are a Nash equilibrium if, for each player i, s_i^* is (at least tied for) player i's best response to the strategies specified for the $n-1$ other players, $(s_1^*, s_2^*, .., s_{i-1}^*, s_{i+1}^* ... s_n^*)$:

$u_i(s_1^*, s_2^*, .., s_{i-1}^*, s_i^*, s_{i+1}^* ... s_n^*) > u_i(s_1^*, s_2^*, .., s_{i-1}^*, s_i, s_{i+1}^* ... s_n^*)$,

for every feasible strategy s_i in S_i.

s_i^* solves

$\max_{s_i \in S_i} u_i(s_1^*, s_2^* ... s_{i-1}^*, s_i, s_{i+1}^*, .. s_n^*)$. At Nash equilibrium $(s_1^*, s_2^*, s_3^* ...., s_n^*)$,

$\max_{s_i \in S_i} u_i(s_1^*, s_2^* ... s_{i-1}^*, s_i^*, s_{i+1}^*, .. s_n^*)$ for each player i.

Results

In the n player normal form game $G = \{S_1 ... S_n; u_1 ... u_n\}$, if iterated elimination of strictly dominated strategies eliminates all but the strategies $(s_1^*, s_2^* ..., s_n^*)$, then these strategies are the unique Nash equilibrium of the game.

In the n player normal form game $G = \{S_1 ... S_n; u_1 ... u_n\}$, if the strategies $(s_1^*, s_2^* ..., s_n^*)$ are a Nash equilibrium, then they survive iterated elimination of strictly dominated strategies.

Suppose player i has K pure strategies; $S_i = \{s_{i1}, s_{i2}, s_{i3}, ...s_{iK}\}$. Then a mixed strategy for player i is a probability distribution $(p_{i1}, p_{i2}, p_{i3}..p_{iK})$ where p_{ik} is the probability that player i will play strategy s_{ik}, for $k = 1, 2, ...K$.

p_{ik} is a probability so $0 \leq p_{ik} \leq 1$ and $\sum_{k=1}^{K} p_{ik} = 1$.